

Introductory Concepts of Probability

STA122 · All 10 Problem Sheets · Full Step-by-Step Solutions · Dr. Adebayo F. A. · January 2025

How to Use These Notes

Every sheet has: **plain-English concept notes** for someone who has never seen the topic, **all original problems** from the worksheet, **SVG visualisations**, and fully worked **step-by-step solutions**.

Sheet 1

Set Theory — union, intersection, complement, Venn diagrams

Sheet 2

Sample Spaces & Basic Probability Laws

Sheet 3

Counting — Permutations & Combinations

Sheet 4

Conditional Probability, Bayes' Theorem, Independence

Sheet 5

Random Variables — Discrete vs Continuous

Sheet 6

PMF, CDF, Bar Graphs

Sheet 7

Expected Value & Variance

Sheet 8

Binomial & Poisson Distributions

Sheet 9

Geometric, Negative Binomial & Hypergeometric

Sheet 10

Uniform, Exponential & Normal Distributions

PREREQUISITES

- Basic algebra — rearranging equations
- Summation notation $\sum$ and product notation $\prod$
- Fractions and percentages
- Basic calculus (integration/differentiation) needed for Sheets 7 & 10

1

Set Theory

THE LANGUAGE OF PROBABILITY

What is a Set?

A **set** is a collection of distinct objects called *elements*. Sets are the foundation of probability because every event is a set of outcomes. We need clean notation to describe "which outcomes are we talking about?"

🌐 **Universal Set U**

All possible elements. Everything lives inside U .

🟪 **Complement $\bar{A}$**

Everything in U NOT in A . If $A = \{1, 2\}$, $U = \{1, 2, 3\}$, then $\bar{A} = \{3\}$.

🟡 **Union $A \cup B$**

Everything in A or B or both. "At least one."

🟦 **Intersection $A \cap B$**

Everything in A and B simultaneously. "Both."

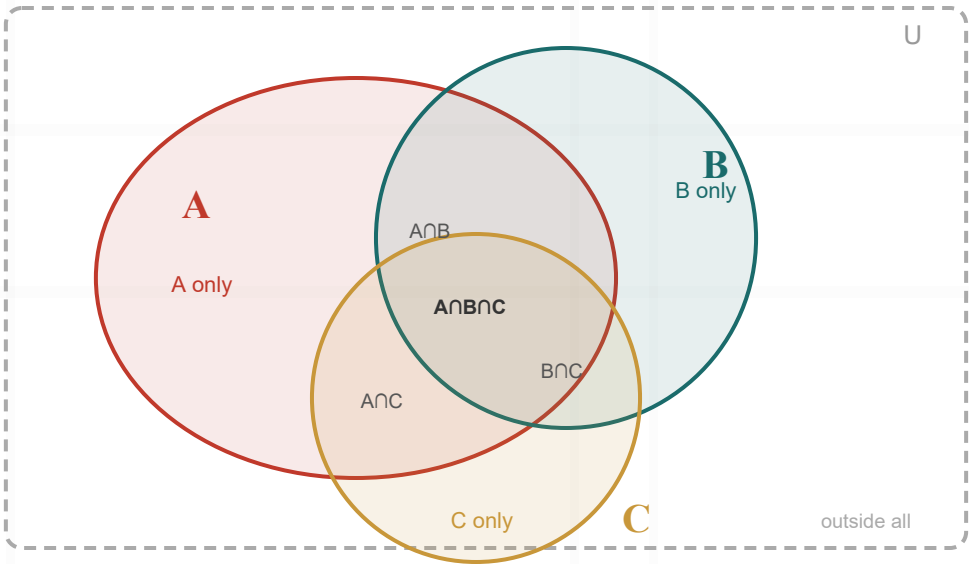
🔴 **$\subset$ Subset $A \subset B$**

Every element of A is also in B .

🟢 **$A - B$ Difference**

$A - B = A \cap \bar{B}$. Elements in A but not in B .

🗺️ **Three-Set Venn Diagram — Regions at a Glance**



ALL PROBLEMS & SOLUTIONS

1

Let $U = \{1, 2, \dots, 10\}$, $A = \{1, 2, \dots, 8\}$, $B = \{2, 3, 4\}$, $C = \{1, 3, 5, 7, 9\}$.

(a) Find $\bar{A}$, $B \cup \bar{C}$, and $A \cap (\bar{B} \cup C)$.

(b) Is it true that $\bar{B} \subset A \cap C$?

Part (a) — Find $\bar{A}$

COMPLEMENT OF A

$\bar{A}$ = elements of U not in A . Since $A = \{1, \dots, 8\}$, everything left is $\bar{A} = \{9, 10\}$.

Find $B \cup \bar{C}$

STEP 1 — FIND $\bar{C}$

$C = \{1, 3, 5, 7, 9\}$ (odd numbers 1-9). So $\bar{C} = \{2, 4, 6, 8, 10\}$ (everything else in U).

STEP 2 — UNION

$B \cup \bar{C} = \{2, 3, 4\} \cup \{2, 4, 6, 8, 10\} = \{2, 3, 4, 6, 8, 10\}$ (combine, no duplicates).

Find $A \cap (\bar{B} \cup C)$

STEP 3 — FIND $\bar{B}$

$B = \{2, 3, 4\}$, so $\bar{B} = \{1, 5, 6, 7, 8, 9, 10\}$.

STEP 4 — FIND $\bar{B} \cup C$

$\{1, 5, 6, 7, 8, 9, 10\} \cup \{1, 3, 5, 7, 9\} = \{1, 3, 5, 6, 7, 8, 9, 10\}$.

STEP 5 — INTERSECT WITH A

$A \cap \{1, 3, 5, 6, 7, 8, 9, 10\} = \{1, \dots, 8\} \cap \{1, 3, 5, 6, 7, 8, 9, 10\} = \{1, 3, 5, 6, 7, 8\}$.

Part (b) — Is $\bar{B} \subset A \cap C$?

FIND $A \cap C$

$A \cap C = \{1, \dots, 8\} \cap \{1, 3, 5, 7, 9\} = \{1, 3, 5, 7\}$.

CHECK:

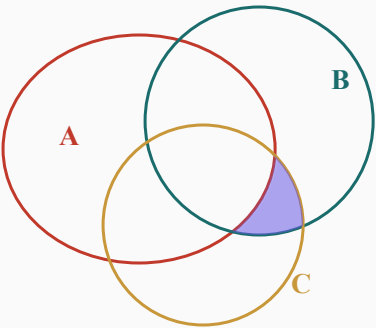
$\bar{B} = \{1, 5, 6, 7, 8, 9, 10\}$. Is $6 \in \{1, 3, 5, 7\}$? NO. So $\bar{B} \not\subset A \cap C$.

✓ $\bar{A} = \{9, 10\}$; $B \cup \bar{C} = \{2, 3, 4, 6, 8, 10\}$; $A \cap (\bar{B} \cup C) = \{1, 3, 5, 6, 7, 8\}$. Part (b): FALSE — $6 \in \bar{B}$ but $6 \notin A \cap C$.

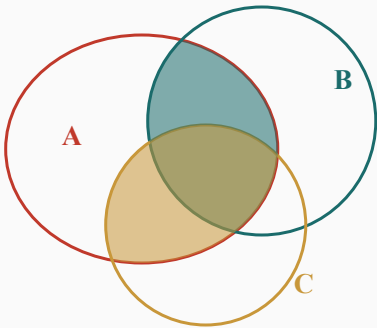
2 Draw Venn diagrams with three sets and shade:
(a) $\bar{A} \cap B \cap C$ (b) $(B \cap A) \cup (A \cap C)$ (c) $A \cup B \cup C$ (d) $(A \cup B \cup C) \cap (\bar{B} \cap C)$

💡 Strategy: identify which of the 7 internal regions (and the outside) satisfy the condition. Work from innermost operations outward.

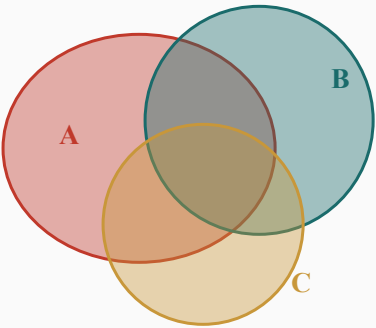
(a) $\bar{A} \cap B \cap C$ — inside B and C but outside A



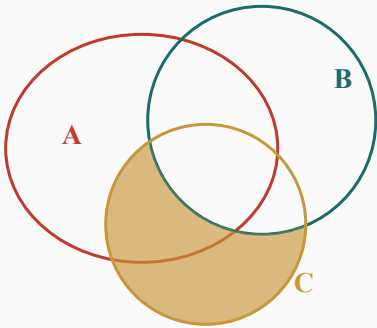
(b) $(A \cap B) \cup (A \cap C) = A$ intersected with B or C



(c) $A \cup B \cup C$ — shade everything inside any circle



(d) $(A \cup B \cup C) \cap (\bar{B} \cap C) = C$ but not B



✓ Each diagram shades exactly the region satisfying the set expression. $(A \cup B \cup C) \cap (\bar{B} \cap C) = \bar{B} \cap C$ (i.e. C minus B), $\bar{B} \cap C \subseteq A \cup B \cup C$ always.
(d) simplifies: since

3 S = all companies on BSE. A =mining, B =turnover>P10m, C =June year-end, D =share price up. Describe in words:

(a) $A \cup B$ (b) $B \cap C$ (c) $B \cap (C \cup A)$ (d) $(B \cap A) \cup (C \cap D)$

💡 Union ($\cup$) = OR. Intersection ($\cap$) = AND. Just substitute the English definitions.

(A) $A \cup B$

All companies that are *in the mining sector OR have annual turnover exceeding P10 million* (or both).

(B) $B \cap C$

Companies that have *annual turnover exceeding P10 million AND a financial year ending in June*.

(C) $B \cap (C \cup A)$

Companies with *turnover above P10 million, AND which either have a June year-end OR are in the mining sector*.

(D) $(B \cap A) \cup (C \cap D)$

Companies that are *(in mining AND high turnover) OR (June year-end AND rising share price)*.

4 Show: (a) $(A \cap \bar{B}) \subset A$ (b) $A - B = A \iff B - A = B$

📄 CORRECTION TO PART (A)

The original sheet states $(A \cap \bar{B}) \subset A \cap B$, which is false (these two sets are always disjoint). The corrected and meaningful statement is $(A \cap \bar{B}) \subset A$.

Part (a): Prove $(A \cap \bar{B}) \subset A$

PROOF (ELEMENT-CHASING METHOD)

Let x be any element such that $x \in A \cap \bar{B}$. By definition of intersection, this means $x \in A$ AND $x \in \bar{B}$. In particular, $x \in A$. Since every element of $A \cap \bar{B}$ is also in A , we have $A \cap \bar{B} \subseteq A$. $\square$

Part (b): Prove $A - B = A \iff B - A = B$

RECALL:

$A - B = A \cap \bar{B}$ and $B - A = B \cap \bar{A}$.

($\Rightarrow$) SUPPOSE $A - B = A$:

This means $A \cap \bar{B} = A$, so every element of A is in $\bar{B}$, i.e. no element of A is in B . Therefore $A \cap B = \emptyset$. Now $B - A = B \cap \bar{A}$. Since $A \cap B = \emptyset$, every element of B lies outside A , so $B \cap \bar{A} = B$, giving $B - A = B$.

($\Leftarrow$) SUPPOSE $B - A = B$:

By the symmetric argument, $A \cap B = \emptyset$, which gives $A - B = A \cap \bar{B} = A$. $\square$

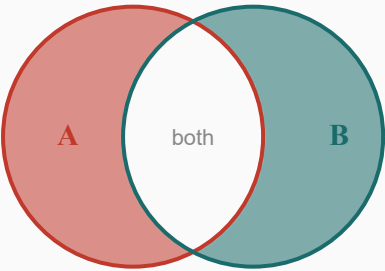
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Depict with Venn diagrams:

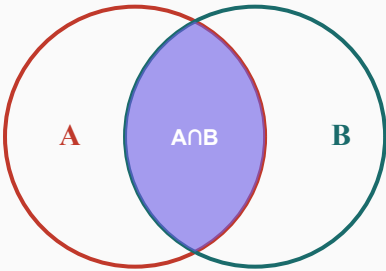
(a) $C = (A \cap \bar{B}) \cup (\bar{A} \cap B)$ (b) $D = (A \cap B) \cup (A \cap B)$ (c) What can you say about C and D ?

💡 $C = (A \cap \bar{B}) \cup (\bar{A} \cap B)$ is called the **symmetric difference** $A \triangle B$ — elements in exactly one of A or B (not both).

(a) $C = A \triangle B$ — A or B but NOT both



(b) $D = A \cap B$ (union with itself is itself)



(C) RELATIONSHIP BETWEEN C AND D:

C (symmetric difference) and D (intersection) are **DISJOINT**

— $C \cap D = \emptyset$. Together they partition $A \cup B$: $C \cup D = A \cup B$. They represent complementary perspectives: C = "exactly one", D = "both".

2

Basic Probability

SAMPLE SPACES, EVENTS & PROBABILITY LAWS

🎯 What is Probability?

Probability is a number in $[0, 1]$ measuring how likely an event is. The **sample space** S lists ALL possible outcomes. An **event** is any subset of S . The three axioms: (1) $P(A) \geq 0$, (2) $P(S) = 1$, (3) for mutually exclusive events, $P(A \cup B) = P(A) + P(B)$.

🎯 Sample Space S

All possible outcomes. E.g., one die:
 $S = \{1, 2, 3, 4, 5, 6\}$.

📌 Event

Any subset of S . "Roll even" = $\{2, 4, 6\}$.

➕ Addition Rule

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Subtract overlap to avoid double-counting.

🔄 Complement Rule

$P(\bar{A}) = 1 - P(A)$.

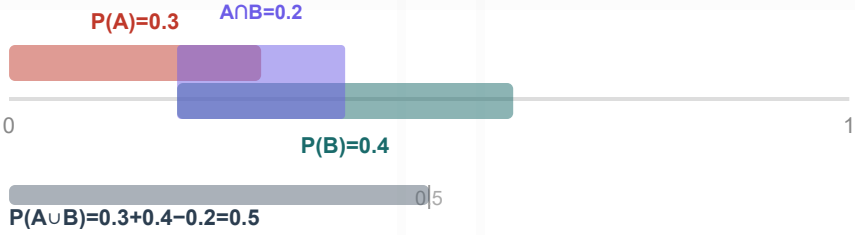
🚫 Mutually Exclusive

$A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B)$.

📊 Classical Probability

$P(A) = \frac{\text{\# outcomes in } A}{\text{total outcomes}}$ when all equally likely.

📊 Addition Rule Visualised — $P(A) = 0.3$, $P(B) = 0.4$, $P(A \cap B) = 0.2$



ALL PROBLEMS & SOLUTIONS

1

A cereal manufacturer packs one of five pictures (A,B,C,D,E) in each box. If you buy **two boxes**, what is the sample space for the two pictures obtained?

💡 Order matters here (first box, second box). Each box independently gets any of the 5 pictures, so the sample space has $5 \times 5 = 25$ elements.

SAMPLE SPACE S :

All ordered pairs (x, y) where $x, y \in \{A, B, C, D, E\}$.

$$S = \{(A, A), (A, B), (A, C), (A, D), (A, E), (B, A), (B, B), (B, C), (B, D), (B, E), \\ (C, A), (C, B), (C, C), (C, D), (C, E), (D, A), (D, B), (D, C), (D, D), (D, E), \\ (E, A), (E, B), (E, C), (E, D), (E, E)\}$$

✓ $|S| = 25$ equally likely outcomes. Five of them (along the diagonal) have both boxes the same picture.

2

A small town has three grocery stores: **Choppies (Ch)**, **Bakgatla (Ba)**, **SaveRite (Sa)**. Four ladies each independently pick a store. Give the sample space and define events:

A: all choose Choppies B: half choose Choppies, half choose Bakgatla C: all stores chosen by at least one lady

SAMPLE SPACE:

Each of 4 ladies picks from $\{Ch, Ba, Sa\}$ independently. Total outcomes = $3^4 = 81$. Each outcome is an ordered 4-tuple e.g. (Ch, Ba, Ch, Sa) .

EVENT A — ALL CHOOSE CHOPPIES:

$A = \{(Ch, Ch, Ch, Ch)\}$ — only 1 outcome.

EVENT B — 2 CHOOSE CHOPPIES, 2 CHOOSE BAKGATLA:

Choose which 2 of the 4 ladies go to Choppies: $\binom{4}{2} = 6$ ways. The other 2 go to Bakgatla. B has 6 outcomes, e.g. $(Ch, Ch, Ba, Ba), (Ch, Ba, Ch, Ba), \dots$

EVENT C — ALL THREE STORES CHOSEN:

At least one lady at each store. Complement: at least one store is empty. By inclusion-exclusion, $|C^c|$ = arrangements missing at least one store = $2^4 + 2^4 + 2^4 - 1 - 1 - 1 = 42$, so $|C| = 81 - 42 = 39$ outcomes.

3

$P(A) = 0.3, P(B) = 0.4, P(A \cap B) = 0.2$. Find:

(a) $P(A \cup B)$ (b) $P(\bar{A})$ (c) $P(\bar{A} \cap \bar{B})$ (d) $P(\text{exactly one of } A \text{ or } B)$

(A) A OR B — ADDITION RULE:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.3 + 0.4 - 0.2 = \mathbf{0.5}$$

(B) A DOES NOT OCCUR:

$$P(\bar{A}) = 1 - P(A) = 1 - 0.3 = \mathbf{0.7}$$

(C) NEITHER A NOR B — DE MORGAN:

$$P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - 0.5 = \mathbf{0.5}$$

(D) EXACTLY ONE:

Exactly one = (A or B) – (A and B) = $P(A \cup B) - P(A \cap B) = 0.5 - 0.2 = \mathbf{0.3}$.

Verify: $P(A \cap \bar{B}) + P(\bar{A} \cap B) = (0.3 - 0.2) + (0.4 - 0.2) = 0.1 + 0.2 = 0.3 \checkmark$

✓ (a) 0.5 | (b) 0.7 | (c) 0.5 | (d) 0.3

4

Show that $P(A \cup \bar{B}) = 1 - [P(B) - P(A \cap B)]$.

STEP 1 — APPLY ADDITION RULE:

$$P(A \cup \bar{B}) = P(A) + P(\bar{B}) - P(A \cap \bar{B})$$

STEP 2 — SUBSTITUTE $P(\bar{B}) = 1 - P(B)$:

$$= P(A) + 1 - P(B) - P(A \cap \bar{B})$$

STEP 3 — USE $P(A \cap \bar{B}) = P(A) - P(A \cap B)$:

(A is split into its overlap with B and its part outside B)

$$\begin{aligned} &= P(A) + 1 - P(B) - (P(A) - P(A \cap B)) \\ &= P(A) + 1 - P(B) - P(A) + P(A \cap B) = 1 - P(B) + P(A \cap B) \\ &= 1 - [P(B) - P(A \cap B)] \quad \square \end{aligned}$$

5

Which are impossible?

(a) $P(A) = 0.7, P(B) = 0.9, P(A \cap B) = 0.4$

(b) $P(A) = 0.2, P(B) = 0.5, P(A \cap B) = 0.25$

(c) $P(A) = 0.3, P(B) = 0.7, P(A \cap B) = 0.1$

(d) $P(A) = 0.8, P(B) = 0.4, P(A \cup B) = 0.7$

💡 Three key rules: (i) $P(A \cap B) \leq \min(P(A), P(B))$, (ii) $P(A \cup B) \leq 1$, (iii) $P(A \cup B) \geq \max(P(A), P(B))$.

(A)

$P(A \cup B) = 0.7 + 0.9 - 0.4 = 1.2 > 1 \times$.

IMPOSSIBLE.

(B)

$P(A \cap B) = 0.25 > P(A) = 0.2 \times$ (intersection can't exceed either set).

IMPOSSIBLE.

(C)

Check: $P(A \cap B) = 0.1 \leq \min(0.3, 0.7) = 0.3 \checkmark$; $P(A \cup B) = 0.3 + 0.7 - 0.1 = 0.9 \leq 1 \checkmark$.
POSSIBLE.

(D)
From Addition Rule: $0.7 = 0.8 + 0.4 - P(A \cap B)$, so $P(A \cap B) = 0.5$. But $0.5 > P(B) = 0.4 \times$.
IMPOSSIBLE.

✓ Impossible: (a), (b), (d). Possible: (c) only.

3

Counting Principles

PERMUTATIONS, COMBINATIONS & CLASSICAL PROBABILITY

Why Do We Need to Count?

When outcomes are equally likely, $P(A) = \frac{\text{\# favourable}}{\text{total outcomes}}$. For large problems, direct listing is impossible — we need smart counting formulas.

Multiplication Principle

Task 1 has m ways, task 2 has n ways $\Rightarrow$ together:
 $m \times n$ ways.

Permutation $P(n, r)$

Ordered arrangements of r from n : $\frac{n!}{(n-r)!}$. $AB \neq BA$.

Combination $\binom{n}{r}$

Unordered selections: $\frac{n!}{r!(n-r)!}$. $\{A,B\}=\{B,A\}$.

Factorial

$n! = n \times (n-1) \times \cdots \times 1$. $5! = 120$. $0! = 1$.

Pigeonhole Principle

If $n + 1$ items go into n boxes, at least one box has
 ≥ 2 items.

Multinomial

Divide n into groups of sizes $r_1, r_2, \dots \therefore \frac{n!}{r_1!r_2!\dots}$

ALL PROBLEMS & SOLUTIONS

1

An 8-digit cellphone number must start with 7, cannot have 0 as the second digit, and no repeated digits are allowed. What is the probability that a randomly generated 8-digit number (starting with 7) satisfies all these conditions?

 Fill the 8 slots one by one, tracking how many valid choices remain at each position.

TOTAL 8-DIGIT NUMBERS STARTING WITH 7, NO REPEATS:

Digit 1 = 7 (1 way). Digits 2–8: fill from remaining 9 digits without repetition.

$$\text{Total} = 1 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 = 9 \times 8!/8 = 9!/1 \div \dots$$

Let's count directly: $9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 = 181,440$

FAVOURABLE (ALSO 2ND DIGIT $\neq 0$):

Digit 1 = 7 (1 way). Digit 2: cannot be 7 (used) or 0 $\rightarrow$ 8 choices. Digit 3: 8 remaining digits. Digit 4: 7. ... Digit 8: 3.

$$\text{Favourable} = 1 \times 8 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 = 8^2 \times 7! = 8 \times 8 \times 5040/\dots$$

$$= 8 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 = 8 \times 20160 = 161,280$$

PROBABILITY:

$$P = \frac{161,280}{181,440} = \frac{8}{9} \approx 0.889$$

✓ $P = \frac{8}{9} \approx 88.9\%$ — Most numbers starting with 7 without repeats also avoid 0 as the 2nd digit.

2

Bridge: 13 cards dealt from 52. Find the probability that a hand contains:

(a) 3 spades, 4 diamonds, 1 heart, 5 clubs

(b) Exactly 3 aces and 4 picture cards (K, Q, J)

💡 Deck: 52 cards, 4 suits of 13. 4 aces total. 12 picture cards (K,Q,J × 4 suits). Total hands = $\binom{52}{13} = 635,013,559,600$.

Part (a)

CHOOSE FROM EACH SUIT INDEPENDENTLY:

$$\begin{aligned} \text{Favourable} &= \binom{13}{3} \binom{13}{4} \binom{13}{1} \binom{13}{5} \\ &= 286 \times 715 \times 13 \times 1287 = 3,421,675,230 \end{aligned}$$

PROBABILITY:

$$P = \frac{3,421,675,230}{635,013,559,600} \approx 0.00539 \approx \mathbf{0.54\%}$$

Part (b)

3 ACES FROM 4, 4 PICTURES FROM 12, REST FROM REMAINING 36:

Remaining cards = $52 - 4 - 12 = 36$ (not aces, not pictures). Need 13 total: $13 - 3 - 4 = 6$ from these 36.

$$\text{Favourable} = \binom{4}{3} \binom{12}{4} \binom{36}{6} = 4 \times 495 \times 1,947,792 = 3,858,609,360$$

PROBABILITY:

$$P = \frac{3,858,609,360}{635,013,559,600} \approx 0.00608 \approx \mathbf{0.61\%}$$

✓ (a) $\approx 0.539\%$ | (b) $\approx 0.608\%$

3

n people in a room. Assuming birth months equally likely:

(a) $P(\text{at least two share a birth month})$

(b) Evaluate for $n = 13$.

💡 Easier to compute $P(\text{all different})$ and subtract from 1 — the complement trick.

TOTAL ARRANGEMENTS OF BIRTH MONTHS:

12^n (each of n people can have any of 12 months).

ALL DIFFERENT MONTHS (ORDERED):

$12 \times 11 \times 10 \times \cdots \times (13 - n) = \frac{12!}{(12-n)!}$ (only possible if $n \leq 12$).

GENERAL FORMULA:

$$P(\text{at least two share}) = 1 - \frac{12!}{(12 - n)! \cdot 12^n}$$

FOR $N=13$:

13 people, only 12 months. By the

PIGEONHOLE PRINCIPLE

, at least two people MUST share a birth month. $P = 1$ (certainty).

✓ $P = 1 - \frac{12!}{(12 - n)! \cdot 12^n}$ for $n \leq 12$. For $n = 13$: $P = 1$ (guaranteed by Pigeonhole).

4

- 75 students divided into tutorial groups of sizes 25, 30, and 20.
- (a) In how many ways can this be done?
- (b) There are two brothers in the class. What is the probability they end up in the same group?

Part (a)

MULTINOMIAL COEFFICIENT:

$$\binom{75}{25} \binom{50}{30} \binom{20}{20} = \frac{75!}{25! 30! 20!}$$

This is an astronomically large number ($\approx 5.7 \times 10^{31}$).

Part (b)

METHOD — FIX ONE BROTHER, FIND PROBABILITY THE OTHER JOINS HIM:

Given brother 1 is placed, there are 74 remaining spots. Brother 2 occupies one of these 74 spots.

If brother 1 is in group of 25: 24 remaining spots in his group, so $P = \frac{24}{74}$.

If brother 1 is in group of 30: $P = \frac{29}{74}$.

If brother 1 is in group of 20: $P = \frac{19}{74}$.

WEIGHT BY PROBABILITY BROTHER 1 IS IN EACH GROUP:

$$\begin{aligned} P(\text{same group}) &= \frac{25}{75} \cdot \frac{24}{74} + \frac{30}{75} \cdot \frac{29}{74} + \frac{20}{75} \cdot \frac{19}{74} \\ &= \frac{1}{75 \times 74} (25 \times 24 + 30 \times 29 + 20 \times 19) = \frac{600 + 870 + 380}{5550} = \frac{1850}{5550} = \frac{1}{3} \end{aligned}$$

✓ (a) $\frac{75!}{25! 30! 20!}$ | (b) $P = \frac{1}{3} \approx 33.3\%$

5

- A student answers a 10-question True/False test. In how many ways can they answer if:
- (a) Exactly half the answers are TRUE and half FALSE?
- (b) No two consecutive answers are the same?

Part (a) — exactly 5 TRUE, 5 FALSE

CHOOSE WHICH 5 QUESTIONS ARE TRUE:

$$\binom{10}{5} = \frac{10!}{5!5!} = 252 \text{ ways}$$

Part (b) — no two consecutive answers the same

PATTERN MUST ALTERNATE:

The sequence must go T,F,T,F,... or F,T,F,T,... — only 2 possible sequences.

VERIFY:

With 10 questions and strict alternation: TFTFTFTFTF or FTFTFTFTFT. Exactly 2 ways.

✓ (a) $\binom{10}{5} = 252$ ways | (b) Only **2** ways (must alternate T and F)

 "Given That..." — Conditional Probability

$P(A|B)$ = "probability of A *given* B has happened." We restrict the universe to just B and ask: what fraction of B is also A ? Bayes' Theorem reverses the conditioning — it answers "given the result, what caused it?"

 Conditional Prob.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \text{ provided } P(B) > 0.$$

 Multiplication Rule

$$P(A \cap B) = P(A|B) \cdot P(B)$$

 Law of Total Prob.

If $B_1, \dots, B_n$ partition S :
$$P(A) = \sum_i P(A|B_i)P(B_i)$$

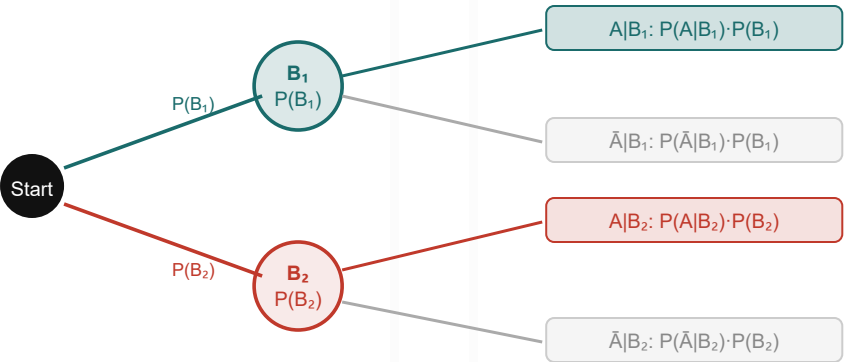
 Bayes' Theorem

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_j P(A|B_j)P(B_j)}$$

 Independence

$A \perp B$ iff $P(A \cap B) = P(A)P(B)$, equivalently
 $P(A|B) = P(A)$.

 Probability Tree — General Bayes Setup



Bayes:
$$P(B_i|A) = \frac{\text{branch to } A \text{ via } B_i}{\text{all branches reaching } A}$$

ALL PROBLEMS & SOLUTIONS

1

P(pass|studied)=0.8, P(pass|not studied)=0.3. 60% of the class studied. Given a randomly chosen student passed, what is P(she studied)?

DEFINE EVENTS:

S =studied ($P(S) = 0.6, P(\bar{S}) = 0.4$). P =passed. $P(P|S) = 0.8, P(P|\bar{S}) = 0.3$.

LAW OF TOTAL PROBABILITY — FIND $P(P)$:

$$P(P) = P(P|S)P(S) + P(P|\bar{S})P(\bar{S}) = 0.8(0.6) + 0.3(0.4) = 0.48 + 0.12 = 0.60$$

BAYES' THEOREM:

$$P(S|P) = \frac{P(P|S) \cdot P(S)}{P(P)} = \frac{0.48}{0.60} = \mathbf{0.80}$$

✓ $P(\text{studied}|\text{passed}) = 0.80$ — 80% chance she studied.

2

Machine A: 40% of output, 10% defective. Machine B: 60% of output, 4% defective.

(a) What proportion of total production is defective?

(b) If a random item is defective, P(from Machine A)?

(A) TOTAL PROBABILITY:

$$P(D) = P(D|A)P(A) + P(D|B)P(B) = 0.10(0.40) + 0.04(0.60) = 0.040 + 0.024 = \mathbf{0.064}$$

So 6.4% of total output is defective.

(B) BAYES:

$$P(A|D) = \frac{P(D|A)P(A)}{P(D)} = \frac{0.040}{0.064} = \mathbf{0.625}$$

✓ (a) 6.4% defective | (b) 62.5% chance it came from Machine A

3

Disease prevalence = 3%. Test: 7% false positive (healthy→+), 2% false negative (diseased→-). Find:
(a) Proportion of diseased correctly diagnosed (b) P(disease|+) (c) P(disease|-) (d) P(incorrect diagnosis)

💡 Let D =diseased. $P(D) = 0.03$, $P(+|\bar{D}) = 0.07$, $P(-|D) = 0.02$. Build a table for 10,000 imaginary people.

	Test +	Test -	Total
Disease	$0.03 \times 0.98 = 0.0294$	$0.03 \times 0.02 = 0.0006$	0.03
No Disease	$0.97 \times 0.07 = 0.0679$	$0.97 \times 0.93 = 0.9021$	0.97
Total	0.0973	0.9027	1.00

(A) P(CORRECT|DISEASE) = P(+|D):

$= 1 - 0.02 = \mathbf{0.98}$ — 98% of diseased people test positive.

(B) P(D|+):

$$= \frac{0.0294}{0.0973} \approx \mathbf{0.302}$$

Only ~30% of positives actually have the disease! (Low base rate effect.)

(C) P(D|-):

$$= \frac{0.0006}{0.9027} \approx \mathbf{0.000665}$$

A negative test is very reassuring.

(D) P(INCORRECT):

False positive + false negative = $0.0679 + 0.0006 = \mathbf{0.0685}$ (6.85%).

✓ (a) 98% | (b) ≈30.2% | (c) ≈0.067% | (d) ≈6.85%

4

$P(A) = 0.6, P(B|A) = 0.3, P(A \cup B) = 0.72$. Are A and B independent, mutually exclusive, or both?

FIND $P(A \cap B)$ VIA MULTIPLICATION RULE:

$$P(A \cap B) = P(B|A) \cdot P(A) = 0.3 \times 0.6 = 0.18$$

FIND $P(B)$ VIA ADDITION RULE:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \Rightarrow 0.72 = 0.6 + P(B) - 0.18 \Rightarrow P(B) = 0.30$$

TEST INDEPENDENCE:

$P(A)P(B) = 0.6 \times 0.3 = 0.18 = P(A \cap B) \checkmark$. So A and B are

INDEPENDENT

.

TEST MUTUAL EXCLUSIVITY:

$P(A \cap B) = 0.18 \neq 0$. So A and B are

NOT

mutually exclusive.

✓ A and B are **independent** but NOT mutually exclusive.

5

If A and B are independent, show that the following pairs are also independent:

(a) A and $\bar{B}$ (b) $\bar{A}$ and $\bar{B}$

💡 Two events X, Y are independent iff $P(X \cap Y) = P(X) \cdot P(Y)$.

Part (a): Show $P(A \cap \bar{B}) = P(A)P(\bar{B})$

STEP 1:

$P(A \cap \bar{B}) = P(A) - P(A \cap B)$ (since $A = (A \cap B) \cup (A \cap \bar{B})$, disjoint).

STEP 2:

$= P(A) - P(A)P(B)$ (using $A \perp B$, so $P(A \cap B) = P(A)P(B)$) $= P(A)(1 - P(B)) = P(A)P(\bar{B})$. $\square$

Part (b): Show $P(\bar{A} \cap \bar{B}) = P(\bar{A})P(\bar{B})$

STEP 1:

$P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B)$

STEP 2:

$= 1 - [P(A) + P(B) - P(A \cap B)] = 1 - P(A) - P(B) + P(A)P(B)$

STEP 3:

$= (1 - P(A))(1 - P(B)) = P(\bar{A})P(\bar{B})$. $\square$

5

Random Variables

DISCRETE & CONTINUOUS; BUILDING DISTRIBUTIONS



What is a Random Variable?

A **random variable (RV)** is a numerical outcome of a random experiment — a function that assigns a number to each outcome. Instead of saying "rolled a 3 then 5", we say $X = \max(3, 5) = 5$. RVs let us do maths on randomness.



Discrete RV

Countable values (often integers). E.g. number of heads in 10 flips.



Continuous RV

Any real value in an interval. E.g. height, time, temperature.



PMF (Discrete)

$p(x) = P(X = x)$. Must satisfy $p(x) \geq 0$ and $\sum p(x) = 1$.

PDF (Continuous)

$f(x) \geq 0$ and $\int_{-\infty}^{\infty} f(x)dx = 1$.
 $P(a \leq X \leq b) = \int_a^b f(x)dx$.

ALL PROBLEMS & SOLUTIONS

1

State whether discrete or continuous:
(a) Time to answer this question (b) Number of words in a random book (c) Max temperature at Sir Seretse Khama Airport (d) Applications before landing a job (e) Proportion of students passing STA122

Variable	Type	Reason
(a) Time to answer	Continuous	Time can be any positive real number of seconds
(b) Words in a book	Discrete	Word count is a whole number (countable)
(c) Max temperature	Continuous	Temperature takes any real value in a range
(d) Applications before job	Discrete	Counts are whole numbers: 1, 2, 3, ...
(e) Proportion passing	Continuous	Any value in $[0, 1]$ is possible

2

A die is rolled twice. Find all possible values and their probabilities for:
(a) M = maximum of both rolls (b) T = sum of both rolls (c) D = first roll minus second roll

💡 Sample space: 36 equally likely pairs (i, j) , $i, j \in \{1, \dots, 6\}$. Count how many pairs give each value.

(a) Maximum M

FORMULA:
 $P(M = m) = \frac{m^2 - (m-1)^2}{36} = \frac{2m-1}{36}$. Reason: $\max \leq m$ means both dice $\leq m$ (m^2 outcomes), subtract $\max \leq m - 1$ ($(m - 1)^2$ outcomes).

m	1	2	3	4	5	6
$P(M = m)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

(b) Sum $T \in \{2, \dots, 12\}$

t	2	3	4	5	6	7	8	9	10	11	12
P	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

(c) Difference $D = X_1 - X_2 \in \{-5, \dots, 5\}$

d	-5	-4	-3	-2	-1	0	1	2	3	4	5
P	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

3

An urn has 6 white, 2 black, 2 red balls. Two balls chosen at random. Win P20 per black ball, lose P10 per white ball. Let X =winnings. Find all values of X and their probabilities.

💡 Total balls = 10. Total pairs = $\binom{10}{2} = 45$. Possible selections: 0,1,2 black balls; then the others are white/red (red has no money effect).

POSSIBLE OUTCOMES BY (BLACK, WHITE, RED) COUNTS IN 2 CHOSEN:

- 0 black, 2 white: $X = -20$. Count: $\binom{2}{0} \binom{6}{2} = 15$.
- 0 black, 1 white, 1 red: $X = -10$. Count: $\binom{2}{0} \binom{6}{1} \binom{2}{1} = 12$.
- 0 black, 0 white, 2 red: $X = 0$. Count: $\binom{2}{0} \binom{6}{0} \binom{2}{2} = 1$.
- 1 black, 1 white: $X = 20 - 10 = 10$. Count: $\binom{2}{1} \binom{6}{1} = 12$.
- 1 black, 0 white, 1 red: $X = 20$. Count: $\binom{2}{1} \binom{6}{0} \binom{2}{1} = 4$.
- 2 black, 0 white: $X = 40$. Count: $\binom{2}{2} = 1$.

x (Pula)	-20	-10	0	10	20	40
$P(X = x)$	$\frac{15}{45} = \frac{1}{3}$	$\frac{12}{45} = \frac{4}{15}$	$\frac{1}{45}$	$\frac{12}{45} = \frac{4}{15}$	$\frac{4}{45}$	$\frac{1}{45}$

✓ Check: $\frac{15+12+1+12+4+1}{45} = \frac{45}{45} = 1$ ✓

4

Multiple-choice test: 5 questions, 3 options each. Student guesses all answers. $X=1$ if pass (≥ 3 correct), $X=0$ if fail. Find values of X and their probabilities.

SETUP:
 Y =number correct. Each answer correct with $p = \frac{1}{3}$, independently. $Y \sim B(5, \frac{1}{3})$. Pass if $Y \geq 3$.

$P(Y \geq 3)$:

$$P(Y = 3) = \binom{5}{3} \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^2 = 10 \times \frac{4}{243} = \frac{40}{243}$$
$$P(Y = 4) = \binom{5}{4} \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^1 = 5 \times \frac{2}{243} = \frac{10}{243}$$
$$P(Y = 5) = \left(\frac{1}{3}\right)^5 = \frac{1}{243}$$
$$P(\text{pass}) = \frac{40 + 10 + 1}{243} = \frac{51}{243} = \frac{17}{81} \approx 0.210$$

X	0 (Fail)	1 (Pass)
$P(X = x)$	$1 - \frac{17}{81} = \frac{64}{81} \approx 0.790$	$\frac{17}{81} \approx 0.210$

✓ $P(\text{pass by guessing}) \approx 21\%$ — guessing has poor odds on 3-option questions.

5

Pool company: completes 2 or 3 contracts/month (equally likely). Receives 1–4 new contracts with $P(1)=0.1$, $P(2)=0.2$, $P(3)=0.5$, $P(4)=0.2$. Starts month with 2 carried forward. X =contracts carried to next month. Find distribution of X .

💡 $X = (\text{carried in} + \text{new contracts received}) - \text{completed}$. Carried in = 2. $C \in \{2, 3\}$ each with prob $\frac{1}{2}$. New contracts $N \in \{1, 2, 3, 4\}$. So $X = 2 + N - C$.

RANGE OF X :
Min: $N = 1, C = 3 \Rightarrow X = 0$. Max: $N = 4, C = 2 \Rightarrow X = 4$. But we don't carry negative, so $X = \max(0, 2 + N - C)$. Since min is 0, $X \in \{0, 1, 2, 3, 4\}$.

$P(X = x)$ BY SUMMING OVER ALL (N, C) GIVING $2 + N - C = x$
Joint: $P(N = n)P(C = c)$, independent.
 $X = 0$: $N = 1, C = 3$: $0.1 \times 0.5 = 0.05$.
 $X = 1$: $N = 1, C = 2$ or $N = 2, C = 3$: $0.1(0.5) + 0.2(0.5) = 0.05 + 0.10 = 0.15$.
 $X = 2$: $N = 2, C = 2$ or $N = 3, C = 3$: $0.2(0.5) + 0.5(0.5) = 0.10 + 0.25 = 0.35$.
 $X = 3$: $N = 3, C = 2$ or $N = 4, C = 3$: $0.5(0.5) + 0.2(0.5) = 0.25 + 0.10 = 0.35$.
 $X = 4$: $N = 4, C = 2$: $0.2(0.5) = 0.10$.

X	0	1	2	3	4
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✓ Sum = $0.05 + 0.15 + 0.35 + 0.35 + 0.10 = 1.00$ ✓

6

PMF & CDF

PROBABILITY MASS FUNCTIONS, CUMULATIVE DISTRIBUTIONS, BAR GRAPHS

PMF and CDF — Two Descriptions of the Same Distribution

The **PMF** gives probability at each exact value: $p(x) = P(X = x)$. The **CDF** gives cumulative probability up to x : $F(x) = P(X \leq x)$. They carry identical information — $p(k) = F(k) - F(k - 1)$ for integer-valued RVs.

✓ Valid PMF

(1) $p(x) \geq 0$ for all x . (2) $\sum_{\text{all } x} p(x) = 1$.

CDF Properties

$F(x)$ is non-decreasing, right-continuous,
 $F(-\infty) = 0, F(\infty) = 1$.

CDF→PMF

$p(k) = F(k) - F(k - 1)$ for integer-valued discrete RVs.

Geometric Series

$\sum_{x=0}^{\infty} ar^x = \frac{a}{1-r}$ for $|r| < 1$. Crucial for checking PMFs with infinite support.

ALL PROBLEMS & SOLUTIONS

1

Which are valid PMFs?

(a) $p(x) = 0.25(0.75)^x, x = 0, 1, 2, \dots$ (b) $p(x) = \frac{1}{5}(2x - 3), x = 0, 1, 2, 3, 4, 5$ (c) $p(x) = \frac{\binom{5}{x}\binom{4}{x-2}}{36}, x = 0, 1, 2$

(A) CHECK NON-NEGATIVITY:

$(0.75)^x > 0$ and $0.25 > 0$, so all terms positive ✓.

CHECK SUM:

$$\sum_{x=0}^{\infty} 0.25(0.75)^x = 0.25 \cdot \frac{1}{1-0.75} = 0.25 \times 4 = 1 \checkmark.$$

VALID PMF ✓

(B) CHECK AT $X=0$:

$$p(0) = \frac{1}{5}(0 - 3) = -0.6 < 0 \text{ X. Probabilities cannot be negative!}$$

NOT A PMF X

(C) EVALUATE EACH VALUE:

$$x = 0: \binom{4}{-2} = 0 \text{ (undefined/zero), so } p(0) = 0.$$

$$x = 1: \binom{4}{-1} = 0, \text{ so } p(1) = 0.$$

$$x = 2: \frac{\binom{5}{2}\binom{4}{0}}{36} = 10 \times 1/36 = 10/36.$$

$$\text{Sum} = 10/36 \neq 1 \text{ X.}$$

NOT A PMF X

✓ Only (a) is a valid PMF.

$$p(x) = \frac{k}{x!} \text{ for } x = 0, 1, 2, 3, 4, \text{ zero otherwise.}$$

2 (a) Find k so that $p(x)$ is a PMF. (b) Find $P(X < 2)$.

(A) SUM MUST EQUAL 1:

$$k \sum_{x=0}^4 \frac{1}{x!} = k \left(\frac{1}{1} + \frac{1}{1} + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} \right) = k \cdot \frac{24 + 24 + 12 + 4 + 1}{24} = k \cdot \frac{65}{24} = 1$$
$$\therefore k = \frac{24}{65} \approx 0.3692$$

(B) $P(X < 2) = P(X = 0) + P(X = 1)$:

$$= \frac{k}{0!} + \frac{k}{1!} = k + k = 2k = \frac{48}{65} \approx 0.738$$

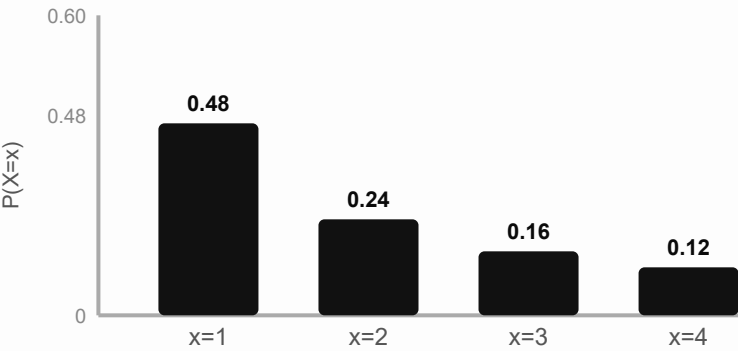
✓ $k = \frac{24}{65}$; $P(X < 2) = \frac{48}{65} \approx 0.738$ </div>

3 $p(x) = \frac{12}{25x}$ for $x = 1, 2, 3, 4$. Plot the bar graph for this PMF.

COMPUTE PROBABILITIES:

$p(1) = \frac{12}{25} = 0.48; p(2) = \frac{12}{50} = 0.24; p(3) = \frac{12}{75} = 0.16; p(4) = \frac{12}{100} = 0.12$. Sum = 1.00 ✓

Bar Graph of PMF



✓ The distribution is decreasing — the largest probability is $x = 1$ (0.48) and smallest $x = 4$ (0.12). This is a valid PMF since all values are positive and sum to 1.

4 CDF: $F(x) = 0$ for $x < 0$; $\frac{1}{2}$ for $0 \leq x < 1$; $\frac{3}{5}$ for $1 \leq x < 2$; $\frac{4}{5}$ for $2 \leq x < 3$; $\frac{9}{10}$ for $3 \leq x < 4$; 1 for $x \geq 4$. Find the PMF.

💡 $p(k) = F(k) - F(k^-)$ = the "jump" at each value. For integers: $p(k) = F(k) - F(k - 1)$.

$p(0) = F(0) - F(-1) = \frac{1}{2} - 0 = \frac{1}{2}$

$p(1) = F(1) - F(0) = \frac{3}{5} - \frac{1}{2} = \frac{6-5}{10} = \frac{1}{10}$

$p(2) = F(2) - F(1) = \frac{4}{5} - \frac{3}{5} = \frac{1}{5} = \frac{2}{10}$

$p(3) = F(3) - F(2) = \frac{9}{10} - \frac{4}{5} = \frac{9-8}{10} = \frac{1}{10}$

$p(4) = F(4) - F(3) = 1 - \frac{9}{10} = \frac{1}{10}$

x	0	1	2	3	4
$p(x)$	$\frac{1}{2}$	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{1}{10}$	$\frac{1}{10}$

✓ Check: $\frac{5+1+2+1+1}{10} = 1$ ✓

7

Expected Value & Variance

CENTRE AND SPREAD OF A RANDOM VARIABLE

⚖ Mean and Spread

$E(X) = \mu$ is the long-run average (centre of mass of the distribution). $\text{Var}(X) = \sigma^2$ measures spread — how far values typically deviate from the mean. Standard deviation $\sigma = \sqrt{\text{Var}(X)}$ is in the same units as X .

🎯 $E(X)$ Discrete

$E(X) = \sum_x x \cdot p(x)$. Weighted average.

🌊 $E(X)$ Continuous

$E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$

📐 Variance Formula

$\text{Var}(X) = E(X^2) - [E(X)]^2$. Shortcut avoids computing deviations.

🔧 $E(X^2)$ Discrete

$E(X^2) = \sum_x x^2 \cdot p(x)$

🔧 $E(X^2)$ Continuous

$E(X^2) = \int x^2 f(x) dx$

📏 Linearity

$E(aX + b) = aE(X) + b$;

$\text{Var}(aX + b) = a^2 \text{Var}(X)$

ALL PROBLEMS & SOLUTIONS

1

$f(x) = 2x$ for $0 \leq x \leq 1$. Find α such that $P(X < \alpha) = 3P(X \geq \alpha)$.

VERIFY f IS A VALID PDF:

$$\int_0^1 2x dx = [x^2]_0^1 = 1 \checkmark$$

SET UP THE EQUATION:

Let $P(X < \alpha) = p$. Then $P(X \geq \alpha) = 1 - p$. Condition: $p = 3(1 - p) \Rightarrow p = 3 - 3p \Rightarrow 4p = 3 \Rightarrow p = \frac{3}{4}$.

SOLVE $P(X < \alpha) = \frac{3}{4}$:

$$\int_0^\alpha 2x dx = [x^2]_0^\alpha = \alpha^2 = \frac{3}{4} \Rightarrow \alpha = \frac{\sqrt{3}}{2} \approx 0.866$$

✓ $\alpha = \frac{\sqrt{3}}{2} \approx 0.866$

2

$f(x) = e^{-x}$ for $x \geq 0$. Find t such that $P(X < t) = 0.5$.

COMPUTE $P(X < t)$:

$$\int_0^t e^{-x} dx = [-e^{-x}]_0^t = 1 - e^{-t}$$

SET EQUAL TO 0.5:

$$1 - e^{-t} = 0.5 \Rightarrow e^{-t} = 0.5 \Rightarrow -t = \ln(0.5) \Rightarrow t = \ln 2 \approx 0.693$$

✓ $t = \ln 2 \approx 0.693$. This is the **median** of the Exponential(1) distribution.

3 $P(X = 1) = 0.3, P(X = 2) = 0.2, P(X = 0) = 3P(X = 3)$. Find $E(X)$ and $\text{Var}(X)$.

FIND MISSING PROBABILITIES:

Let $P(X = 3) = p, P(X = 0) = 3p$. All sum to 1: $3p + 0.3 + 0.2 + p = 1 \Rightarrow 4p = 0.5 \Rightarrow p = 0.125$. So $P(X = 0) = 0.375, P(X = 3) = 0.125$.

x	0	1	2	3
$p(x)$	0.375	0.300	0.200	0.125
$x \cdot p(x)$	0	0.300	0.400	0.375
$x^2 \cdot p(x)$	0	0.300	0.800	1.125

$E(X) = 0 + 0.3 + 0.4 + 0.375 = \mathbf{1.075}$

$E(X^2) = 0 + 0.3 + 0.8 + 1.125 = \mathbf{2.225}$

$\text{Var}(X) = E(X^2) - [E(X)]^2 = 2.225 - (1.075)^2 = 2.225 - 1.155625 = \mathbf{1.069}$

✓ $E(X) = 1.075$; $\text{Var}(X) \approx 1.069$; $\sigma \approx 1.034$

4 Pool company (same setup as Sheet 5 Q5). Find $E(X)$ and $\text{Var}(X)$ for X = contracts carried forward.

💡 Use the distribution from Sheet 5 Q5: $P(X = 0) = 0.05$, $P(X = 1) = 0.15$, $P(X = 2) = 0.35$, $P(X = 3) = 0.35$, $P(X = 4) = 0.10$.

x	0	1	2	3	4
$p(x)$	0.05	0.15	0.35	0.35	0.10
$xp(x)$	0	0.15	0.70	1.05	0.40
$x^2p(x)$	0	0.15	1.40	3.15	1.60

$E(X) = 0 + 0.15 + 0.70 + 1.05 + 0.40 = \mathbf{2.30}$

$E(X^2) = 0 + 0.15 + 1.40 + 3.15 + 1.60 = \mathbf{6.30}$

$\text{Var}(X) = 6.30 - (2.30)^2 = 6.30 - 5.29 = \mathbf{1.01}$

✓ $E(X) = 2.30$ contracts; $\text{Var}(X) = 1.01$; $\sigma \approx 1.005$

5 $f(x) = \frac{3}{4}(kx - x^2)$ for $0 \leq x \leq 2$. Find $k, E(X)$ and $\text{Var}(X)$.

STEP 1 – FIND K

$$\int_0^2 \frac{3}{4}(kx - x^2) dx = 1 \Rightarrow \frac{3}{4} \left[k \frac{x^2}{2} - \frac{x^3}{3} \right]_0^2 = \frac{3}{4} \left(2k - \frac{8}{3} \right) = 1$$

$$2k - \frac{8}{3} = \frac{4}{3} \Rightarrow 2k = 4 \Rightarrow k = 2$$

STEP 2 – $E(X)$ WITH $F(X) = \frac{3}{4}(2X - X^2)$:

$$\begin{aligned} E(X) &= \int_0^2 x \cdot \frac{3}{4}(2x - x^2) dx = \frac{3}{4} \int_0^2 (2x^2 - x^3) dx \\ &= \frac{3}{4} \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_0^2 = \frac{3}{4} \left(\frac{16}{3} - 4 \right) = \frac{3}{4} \cdot \frac{4}{3} = 1 \end{aligned}$$

STEP 3 – $E(X^2)$:

$$\begin{aligned} E(X^2) &= \frac{3}{4} \int_0^2 x^2(2x - x^2) dx = \frac{3}{4} \int_0^2 (2x^3 - x^4) dx \\ &= \frac{3}{4} \left[\frac{x^4}{2} - \frac{x^5}{5} \right]_0^2 = \frac{3}{4} \left(8 - \frac{32}{5} \right) = \frac{3}{4} \cdot \frac{8}{5} = \frac{6}{5} \end{aligned}$$

STEP 4 – $\text{VAR}(X)$:

$$= \frac{6}{5} - 1^2 = \frac{1}{5} = 0.2$$

✓ $k = 2$; $E(X) = 1$; $\text{Var}(X) = \frac{1}{5} = 0.2$; $\sigma = \frac{1}{\sqrt{5}} \approx 0.447$

8

Binomial & Poisson Distributions

TWO INDISPENSABLE NAMED DISTRIBUTIONS

Named Distributions — Ready-Made Formulas

Many real scenarios share the same mathematical structure. The **Binomial** counts successes in n independent trials. The **Poisson** counts rare events in a fixed time/space window. Both save enormous calculation effort.

Binomial $B(n, p)$

$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$. Mean $\mu = np$. Var $= np(1 - p)$.

Poisson $\text{Pois}(\lambda)$

$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$. Mean = Var = λ .

Binomial Conditions

Fixed n ; binary outcome; constant p ; trials independent.

Poisson Conditions

Events rare, random, independent; constant rate λ ; can scale: λ over time t is λt .

Binomial $B(15, 0.3)$ — probability distribution



Mean = $np = 15 \times 0.3 = 4.5$. Distribution is roughly bell-shaped around $k = 4$ or 5 .

ALL PROBLEMS & SOLUTIONS

1

True-false test, 5 questions. Need ≥ 4 correct to pass. Each answer guessed independently. P(pass)?

$X \sim B(5, 0.5)$. NEED $P(X \geq 4)$:

$$P(X = 4) = \binom{5}{4} (0.5)^5 = 5 \times \frac{1}{32} = \frac{5}{32}$$

$$P(X = 5) = \binom{5}{5} (0.5)^5 = \frac{1}{32}$$

$$P(X \geq 4) = \frac{6}{32} = \frac{3}{16} = 0.1875$$

✓ $P(\text{pass}) = \frac{3}{16} = 18.75\%$

2

Engine fails independently with $P=0.1$. Plane succeeds if $\leq$ half engines fail. Compare 2-engine vs 4-engine planes.

2-ENGINE (SUCCEED IF 0 OR 1 FAIL), $X \sim B(2, 0.1)$:

$$P(\text{success}) = P(X \leq 1) = (0.9)^2 + 2(0.1)(0.9) = 0.81 + 0.18 = \mathbf{0.99}$$

4-ENGINE (SUCCEED IF 0,1, OR 2 FAIL), $X \sim B(4, 0.1)$:

$$P(X = 0) = (0.9)^4 = 0.6561$$

$$P(X = 1) = 4(0.1)(0.9)^3 = 0.2916$$

$$P(X = 2) = 6(0.01)(0.81) = 0.0486$$

$$P(\text{success}) = 0.6561 + 0.2916 + 0.0486 = \mathbf{0.9963}$$

✓ 2-engine: 0.99 | 4-engine: 0.9963. The 4-engine plane is slightly more reliable.

3

Right-turn lane holds 3 cars. 30% of 15 approaching cars want to turn right. P(lane insufficient = more than 3 want to turn right)?

$X \sim B(15, 0.3)$. LANE INSUFFICIENT IF $X > 3$, I.E., $P(X \geq 4) = 1 - P(X \leq 3)$:

$$P(X = 0) = (0.7)^{15} = 0.004748$$

$$P(X = 1) = 15(0.3)(0.7)^{14} = 0.030518$$

$$P(X = 2) = \binom{15}{2} (0.3)^2 (0.7)^{13} = 105 \times 0.09 \times 0.009612 = 0.09161$$

$$P(X = 3) = \binom{15}{3} (0.3)^3 (0.7)^{12} = 455 \times 0.027 \times 0.013737 = 0.16969$$

$$P(X \leq 3) = 0.00475 + 0.03052 + 0.09161 + 0.16969 = 0.2966$$

$$P(X \geq 4) = 1 - 0.2966 = \mathbf{0.7034}$$

✓ $P(\text{lane insufficient}) \approx 70.3\%$ — the lane is very likely to be too small!

4

Ice-cream sales: Poisson, $\lambda = 10$ /hour.

(a) P(at least one sale in first hour)?

(b) Minimum stock to be 95% sure of not running out in any given hour?

(A) $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-10} \approx 1 - 0.0000454 \approx \mathbf{0.99995}$

(B) FIND SMALLEST K WITH $P(X \leq K) \geq 0.95$:

Using Poisson CDF with $\lambda = 10$:

k	$P(X \leq k)$	$\geq 0.95?$
13	0.8645	No
14	0.9165	No
15	0.9513	Yes ✓

✓ (a) $\approx 99.995\%$ | (b) Stock at least **15 ice-creams** .

5

Pump fails on average once per 500 hours. P(fails exactly 3 times in 2000 hours)?

RATE OVER 2000 HOURS:

$\lambda = \frac{2000}{500} = 4$ failures expected.

POISSON:

$$P(X = 3) = \frac{e^{-4} \cdot 4^3}{3!} = \frac{e^{-4} \times 64}{6} \approx \frac{0.01832 \times 64}{6} = \frac{1.172}{6} \approx \mathbf{0.1954}$$

✓ $P(X = 3) = \frac{64e^{-4}}{6} \approx 19.5\%$

9

Geometric, Negative Binomial & Hypergeometric

THREE MORE DISTRIBUTIONS



Three More Distributions

Geometric: trials until first success (e.g. first head). **Negative Binomial:** trials until the r -th success. **Hypergeometric:** sampling *without replacement* from a finite population — like drawing from a deck without shuffling back.



Geometric $G(p)$

$P(X = k) = (1 - p)^{k-1}p, k = 1, 2, \dots$ Mean = $\frac{1}{p}$
. Memoryless.



Negative Binomial $NB(r, p)$

$P(X = k) = \binom{k-1}{r-1}p^r(1 - p)^{k-r}, k = r, r + 1, \dots$
Mean = $\frac{r}{p}$.



Hypergeometric

N total, K special, sample n :
 $P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$

 Key Distinction

Binomial = *with replacement* (or large population).
Hypergeometric = *without replacement* from small population.

ALL PROBLEMS & SOLUTIONS

1

TV manufacturer quality control: sample 10, reject if ≥ 1 defective. Find rejection probabilities for defect rates 1%, 10%, 30%. Then compare with: sample 20, reject if ≥ 2 defective. Which is better?

Plan A: $n = 10$, reject if $X \geq 1$

$$P(\text{REJECT}) = 1 - P(X = 0) = 1 - (1 - P)^{10};$$
$$p = 0.01: 1 - (0.99)^{10} = 1 - 0.9044 = \mathbf{0.0956}$$
$$p = 0.10: 1 - (0.90)^{10} = 1 - 0.3487 = \mathbf{0.6513}$$
$$p = 0.30: 1 - (0.70)^{10} = 1 - 0.0282 = \mathbf{0.9718}$$

Plan B: $n = 20$, reject if $X \geq 2$

$$P(\text{REJECT}) = 1 - (1 - P)^{20} - 20P(1 - P)^{19};$$
$$p = 0.01: 1 - (0.99)^{20} - 20(0.01)(0.99)^{19} = 1 - 0.8179 - 0.1652 = \mathbf{0.0169}$$
$$p = 0.10: 1 - (0.90)^{20} - 20(0.1)(0.9)^{19} = 1 - 0.1216 - 0.2702 = \mathbf{0.6082}$$
$$p = 0.30: 1 - (0.70)^{20} - 20(0.3)(0.7)^{19} = 1 - 0.0008 - 0.0068 = \mathbf{0.9924}$$

Defect rate	Plan A: $n = 10$, reject if ≥ 1	Plan B: $n = 20$, reject if ≥ 2
1% (good stock)	9.56% (too high!)	1.69% (better)
10%	65.1%	60.8%
30% (bad stock)	97.2%	99.2% (better)

✓ Plan B is better: lower false rejection of good stock (1.7% vs 9.6%) AND higher true rejection of bad stock (99.2% vs 97.2%).

2

Electric motors: average demand 8/week. Delivery takes 1 week. How low can stock fall before ordering, to be 95% sure of meeting demand during the wait?

DEMAND DURING WAIT $\sim$ POISSON ($\lambda = 8$).
Need smallest k with $P(X \leq k) \geq 0.95$.

k	$P(X \leq k)$	$\geq 0.95?$
10	0.8159	No
11	0.8881	No
12	0.9362	No
13	0.9658	Yes ✓

✓ Allow stock to fall to at minimum **13 motors** before ordering (to be 95% sure demand during delivery is met).

3

Fair die tossed repeatedly. Find:
(a) P(6 appears for the first time on the 10th toss)
(b) P(6 appears for the 6th time on the 10th toss)

(a) Geometric — first success on toss 10

$X \sim G(P = \frac{1}{6})$, NEED $P(X = 10)$:

$$P(X = 10) = \left(\frac{5}{6}\right)^9 \cdot \frac{1}{6} = \frac{5^9}{6^{10}} = \frac{1953125}{60466176} \approx \mathbf{0.03230}$$

(b) Negative Binomial — 6th success on toss 10

NEED EXACTLY 5 SIXES IN TOSSES 1–9, THEN A 6 ON TOSS 10:

$$\begin{aligned} P &= \binom{9}{5} \left(\frac{1}{6}\right)^5 \left(\frac{5}{6}\right)^4 \cdot \frac{1}{6} = \binom{9}{5} \left(\frac{1}{6}\right)^6 \left(\frac{5}{6}\right)^4 \\ &= 126 \times \frac{625}{6^{10}} = \frac{78750}{60466176} \approx \mathbf{0.001302} \end{aligned}$$

✓ (a) $\approx 3.23\%$ | (b) $\approx 0.130\%$

4

Presidential candidate needs 50 signatures. P(any random delegate signs) = 0.15. Find P(need to approach exactly 300 delegates to get the 50th signature).

💡 This is a Negative Binomial problem: we want the 50th success ($r = 50$) to occur on exactly the 300th trial ($k = 300$), with $p = 0.15$.

EXACTLY 49 SUCCESSES IN FIRST 299 TRIALS, THEN A SUCCESS ON TRIAL 300:

$$P(X = 300) = \binom{299}{49} (0.15)^{50} (0.85)^{250}$$

THIS IS A VERY SMALL NUMBER

(approximation via Normal or direct computation): Using logarithms: $\ln P = \ln \binom{299}{49} + 50 \ln(0.15) + 250 \ln(0.85)$. This evaluates to approximately **0.0000235** or about 0.00235%. The exact computation requires software.

✓ $P = \binom{299}{49} (0.15)^{50} (0.85)^{250} \approx 0.0000235$ — an extremely small probability.

5

10 milk cartons, 3 sour. Choose 4 at random. P(at least 2 are sour)?

💡 This is Hypergeometric: $N = 10$, $K = 3$ (sour), $n = 4$ (chosen). No replacement.

TOTAL WAYS TO CHOOSE 4 FROM 10:

$$\binom{10}{4} = 210$$

$P(X = 2)$:

$$\frac{\binom{3}{2} \binom{7}{2}}{\binom{10}{4}} = \frac{3 \times 21}{210} = \frac{63}{210}$$

$P(X = 3)$:

$$\frac{\binom{3}{3} \binom{7}{1}}{\binom{10}{4}} = \frac{1 \times 7}{210} = \frac{7}{210}$$

$$P(X \geq 2) = \frac{63 + 7}{210} = \frac{70}{210} = \frac{1}{3}$$

✓ $P(\text{at least 2 sour}) = \frac{1}{3} \approx 33.3\%$

10

Continuous Distributions

UNIFORM, EXPONENTIAL & NORMAL

Continuous Random Variables

For continuous RVs, $P(X = \text{any exact value}) = 0$. Instead we compute areas under the density curve: $P(a \leq X \leq b) = \int_a^b f(x) dx$. Three fundamental continuous distributions appear constantly in statistics and real-world modelling.

Uniform $U(a, b)$

$f(x) = \frac{1}{b-a}$ on $[a, b]$. Equally likely everywhere.
Mean = $\frac{a+b}{2}$.

Exponential $\text{Exp}(\lambda)$

$f(x) = \lambda e^{-\lambda x}$, $x \geq 0$. Waiting times. Memoryless:
 $P(X > s + t | X > s) = P(X > t)$.

Normal $N(\mu, \sigma^2)$

Bell curve. Standardise: $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$. Use Z-tables for areas.

Exponential Probabilities

$P(X \leq t) = 1 - e^{-\lambda t}$; $P(X > t) = e^{-\lambda t}$

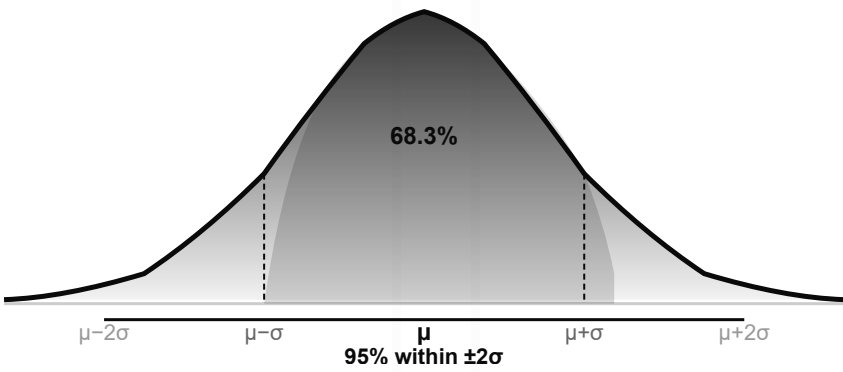
68-95-99.7 Rule

For $N(\mu, \sigma^2)$: 68% within $\pm\sigma$, 95% within $\pm 2\sigma$, 99.7% within $\pm 3\sigma$.

Sum of Normals

If $X_i \sim N(\mu, \sigma^2)$ independently, then
 $\sum X_i \sim N(n\mu, n\sigma^2)$.

Normal Distribution $N(\mu, \sigma^2)$ and the 68-95-99.7 Rule



ALL PROBLEMS & SOLUTIONS

1

Radio announces time every 15 minutes. You wake at a uniformly random time. P(wait less than 5 minutes to hear the time)?

$X \sim U(0, 15)$, $f(x) = \frac{1}{15}$:

$$P(X < 5) = \int_0^5 \frac{1}{15} dx = \frac{5}{15} = \frac{1}{3} \approx 0.333$$

</div>

✓ $P(\text{wait} < 5 \text{ min}) = \frac{1}{3} \approx 33.3\%$

2

Electronic device: $X \sim \text{Exp}(\lambda = 1/1000)$.

(a) P(lasts less than 1000 hours)? (b) P(lasts more than 1200 hours)? (c) 3 devices chosen. P(exactly one lasts less than 800 hours)?

💡 Key: $P(X \leq t) = 1 - e^{-\lambda t}$ and $P(X > t) = e^{-\lambda t}$ with $\lambda = 0.001$.

(A) $P(X < 1000) = 1 - E^{-0.001 \times 1000} = 1 - E^{-1} \approx 1 - 0.3679 = \mathbf{0.6321}$

(B) $P(X > 1200) = E^{-0.001 \times 1200} = E^{-1.2} \approx \mathbf{0.3012}$

(C) LET $P = P(X < 800) = 1 - E^{-0.8} \approx 0.5507$. THREE INDEPENDENT DEVICES, $Y \sim B(3, P)$:

$$P(Y = 1) = \binom{3}{1} (0.5507)(1 - 0.5507)^2 = 3 \times 0.5507 \times (0.4493)^2 \\ = 3 \times 0.5507 \times 0.2019 \approx \mathbf{0.3334}$$

✓ (a) 63.2% | (b) 30.1% | (c) 33.3%

3

Customers arrive at rate 30/hour at a restaurant. One minute has already passed since last customer. P(at least another minute before next)?

💡 Inter-arrival times $\sim \text{Exp}(\lambda)$. Rate = 30/hour = 0.5/minute, so $\lambda = 0.5$.

NEED $P(T > 2 | T > 1)$.

Use the

MEMORYLESS PROPERTY

of the exponential: $P(T > s + t | T > s) = P(T > t)$.

THEREFORE:

$$P(T > 2 | T > 1) = P(T > 1) = e^{-\lambda \cdot 1} = e^{-0.5} \approx \mathbf{0.6065}$$

The fact that 1 minute has already passed is irrelevant — the exponential "forgets" the past!

✓ $P \approx 60.65\%$. The memoryless property means waiting history doesn't change future probability.

4

$X \sim N(3, 4)$ (mean=3, variance=4, so $\sigma = 2$). Evaluate:

(a) $P(X < 4)$ (b) $P(|X| < 6)$ (c) Find a such that $\$P(X$

💡 Standardise: $Z = \frac{X-3}{2} \sim N(0, 1)$. Then use Z-table values: $\Phi(z) = P(Z$

(A) $P(X < 4)$:

$$Z = \frac{4-3}{2} = 0.5. P(Z < 0.5) = \Phi(0.5) = \mathbf{0.6915}$$

(B) $\$P(|X| < 6) = P(-6 \leq X \leq 6) = P\left(-\frac{6-3}{2} \leq Z \leq \frac{6-3}{2}\right) = P(-1.5 \leq Z \leq 1.5)$.

$\$P(-1.5 \leq Z \leq 1.5) = \Phi(1.5) - \Phi(-1.5) \approx 0.9332 - 0.0668 = 0.8664$ (NOTE: $\Phi(-1.5) \approx 0$ SINCE -1.5 IS EXTREMELY FAR IN THE LEFT TAIL.)

(C) $\$P(X \leq a) = 0.80$. NEED $\Phi\left(\frac{a-3}{2}\right) = 0.80$. FROM Z-TABLES: $Z = 0.842$ GIVES $\Phi(0.842) = 0.80$.

SO $\frac{a-3}{2} = 0.842 \Rightarrow a = 3 + 2(0.842) = 3 + 1.684 = \mathbf{4.684}$

✓ (a) 0.6915 | (b) 0.8664 | (c) $a \approx 4.684$

5

$P(X < 10) = 0.8413$ and $\text{Var}(X) = \sigma^2 = 16$. Find the mean μ .

$\Sigma = \sqrt{16} = 4$. STANDARDISE:

$$P(X < 10) = P\left(Z < \frac{10 - \mu}{4}\right) = 0.8413$$

FROM Z-TABLES:

$\Phi(1) = 0.8413$. So $\frac{10-\mu}{4} = 1$.

SOLVE:

$10 - \mu = 4 \Rightarrow \mu = 6$

✓ $\mu = 6$

6

Lift maximum: 10 passengers OR 600kg. Passenger masses $\sim N(65, 15^2)$ kg. How often will 10 random passengers exceed 600kg combined?

TOTAL MASS $T = X_1 + \dots + X_{10}$, EACH $N(65, 225)$ INDEPENDENTLY:

$$T \sim N(10 \times 65, 10 \times 225) = N(650, 2250)$$

Mean = 650 kg, Std dev $\sigma_T = \sqrt{2250} \approx 47.43$ kg.

STANDARDISE:

$$\begin{aligned} P(T > 600) &= P\left(Z > \frac{600 - 650}{47.43}\right) = P(Z > -1.054) \\ &= 1 - \Phi(-1.054) = \Phi(1.054) \approx \mathbf{0.854} \end{aligned}$$

 INTERPRETATION

The mean mass of 10 people is $10 \times 65 = 650$ kg, which already exceeds the 600kg limit! The limit is dangerously low for a population with mean 65kg and std dev 15kg. About 85.4% of random groups of 10 will exceed the limit.

✓ $P(\text{exceed 600kg}) \approx \mathbf{85.4\%}$ — the stated limit is regularly exceeded.